Engineering Memory in Quantum Reservoir Computing through Dissipator Geometry

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Quantum reservoir computing processes temporal data through fixed quantum dynamics, but useful operation requires controlled forgetting. Dissipation is commonly tuned through an overall rate; here we show that its organization is an architectural degree of freedom. We define dissipator geometry by jump family and rate profile: which system combinations couple to the environment and how assigned coupling weight is distributed. In paired comparisons of continuously driven spin reservoirs, we hold fixed the coherent processor, input, Pauli readout, data splits, and operator-weight budget. In the principal comparison, equal-phase collective relaxation produces a pronounced short-term memory advantage over uniform local relaxation, yet no continuously driven design leads every tested task. This shared-channel ordering recurs in separate comparisons with matched mean expected jump activity, a matched midpoint Liouvillian gap, bounded independent operating-point selection, and reset-based encoding, establishing an architecture-level result for the channel studied here. Broader family rankings remain conditional on the fixed-weight protocol. Dissipator geometry can thus systematically shape what a reservoir remembers and computes.

1 Introduction

Reservoir computing processes temporal information with a fixed dynamical system and trains only a simple readout [1, 2]. Quantum reservoir computing (QRC) applies this principle to quantum dynamics [3], making complex evolution useful without optimizing every component of the processor. For time-dependent tasks, however, a reservoir must balance memory and forgetting: recent inputs must remain accessible, while obsolete information and the initial state must lose influence. In open quantum systems, coupling to the environment can provide this controlled forgetting, and dissipative dynamics can realize nonlinear fading-memory input–output maps [4]. Dissipation can therefore be part of the computation rather than merely a source of error.

Previous work has established several ways in which non-unitary dynamics can support QRC. Noise, damping, optical absorption, and measurement back-action can generate or regulate temporal memory [5, 6, 7, 8, 9, 10], while feedback, finite-history reinitialization, non-Markovian embeddings, and optical feedback provide additional mechanisms [11, 12, 13, 14]. Studies of coherence and mid-circuit measurement further delimit practically accessible operating regimes [15, 16]. In particular, Sannia *et al.* introduced a continuously driven spin reservoir with uniform local relaxation and showed that performance can be improved by tuning the damping strength [6]. Related work varies scalar controls such as drive, coupling, input duration, and damping rate [17, 18]. These studies tune the strength of a prescribed dissipator; here we vary its generator-level organization. Known dark or slowly relaxing combinations under shared decay [19, 20] motivate—but do not constitute—our result: a controlled, input-driven, Pauli-visible recent-input memory advantage of the equal-phase shared channel over uniform local loss.

We address this question through *dissipator geometry*, defined by two operational coordinates. The jump family specifies which combinations of the quantum system couple directly to the environment; the rate profile specifies how the available coupling weight is distributed within that family. Independent local relaxation and a

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shared collective channel, for example, expose different combinations of the same sitewise lowering actions: the former couples separate site directions, whereas the latter couples one weighted collective direction. This distinction is defined at the generator level, so equivalent remixings of a jump-operator list are not counted as different architectures. Section 2 formalizes the design space, gives the local–collective intuition, and defines the normalization used for controlled comparison.

We study this design space in continuously driven spin reservoirs based on Ref. [6]. In the principal structural and static-profile comparisons, paired reservoirs share the Hamiltonian instance, input sequence, encoding, Pauli observables, ridge-readout protocol, and data splits, while jump family or rate profile is varied under the declared assigned-weight control. Six jump families are evaluated on short-term memory, NARMA-10, parity, and Mackey–Glass forecasting. Separate STM studies compare several within-family profiles. We additionally test when the exact-expectation STM ordering becomes statistically resolvable under finite measurement budgets.

Within the common fixed-$\mathcal{B}$ protocol, the tested jump families produce distinct task profiles, and no continuously driven design leads every benchmark. We therefore use task specialization here in a protocol-specific sense, not as a ranking after separate task-wise optimization of every family. Against this task-dependent background, one result is especially robust. Compared with independent local losses, the equal-phase shared relaxation channel studied here makes more recent input history accessible to the readout. This collective-memory advantage recurs throughout the tested size sweep, across Hamiltonians that change the interaction and drive axes, interaction form, or connectivity, and after strict washout removes dependence on the tested initial states. It also survives matching realized jump activity or a representative relaxation timescale, as well as independent operating-point selection. Matched rank-one direction interventions further isolate this geometric coordinate. The primary phase-path study and a separately generated, real sign-balanced replication at a second finite size both show that accessible memory depends on which collective direction is coupled when rank, spectrum, trace, sitewise diagonal weights, coefficient magnitudes, and assigned operator weight are fixed (Appendix F).

The effect also transfers to an input-by-reset reservoir, where each input resets one qubit before autonomous evolution. Fresh pairs reproduce the favorable ordering in memory and nonlinear temporal prediction. Together with the size, Hamiltonian, washout, and operating-point checks, this supports robustness across the tested spin-reservoir architectures rather than one size, input mechanism, or scalar setting.

Dynamical diagnostics support a shared-channel interpretation, but the claim is architectural rather than an asymptotic law or an identification of one uniquely causal microscopic ingredient; fixed assigned weight is not physical equivalence. Section 2 defines the model and design space, Section 3 presents the computational results, Section 4 examines the local–collective dynamics, and Section 5 discusses scope and implications.

2 Reservoir model and controlled dissipator design

2.1 The fixed driven reservoir

We keep the coherent reservoir fixed and change only its coupling to the environment. The reservoir is the continuously driven network of N interacting qubits introduced in Ref. [6]. During the kth input interval, a scalar value $s_k \in [0, 1]$ is held constant for a duration Δt, and the qubits evolve under

$$H(s_k) = \sum_{i<j} J_{ij}\sigma_i^x\sigma_j^x + h\sum_i \sigma_i^z + h(1+s_k)\sum_i \sigma_i^x. \tag{1}$$

The J_{ij} term couples the qubits, the z field sets their static energy scale, and the final term writes s_k into the transverse drive. Each reservoir instance uses one sampled set of couplings J_{ij}. This fixes how the input enters and how information moves before the environmental organization is varied.

2.2 Environmental dynamics and reference relaxation

The Hamiltonian describes isolated evolution, while the environment adds a second contribution. For the memoryless model used here, the density matrix obeys the Lindblad master equation [21, 22, 23],

$$\begin{aligned}\dot{\rho} = \mathcal{L}_s(\rho) &= -i[H(s), \rho] + \sum_k \mathcal{D}[L_k](\rho), \\ \mathcal{D}[L](\rho) &= L\rho L^\dagger - \tfrac{1}{2}\{L^\dagger L, \rho\}.\end{aligned} \tag{2}$$

We use Eq. (2) as a target engineered GKLS generator, not as the generic weak-coupling master equation of an arbitrary common bath. The commutator gives the coherent motion. Each $\mathcal{D}[L_k]$ describes one environmental action on average. The operator L_k identifies the transition and its coefficient sets the strength. Both parts act continuously. Because the dissipative term is quadratic in its jump operator,

$$\mathcal{D}[aL] = |a|^2 \mathcal{D}[L],$$

a coefficient $\sqrt{\gamma}$ makes γ the corresponding rate scale.

The standard model gives each qubit its own relaxation channel. Following Sannia *et al.* [6], we write

$$L_i = \sqrt{\gamma}\,\sigma_i^-, \qquad i = 1, \ldots, N. \tag{3}$$

The lowering operator σ_i^- maps the excited state of qubit i to its ground state. One operator per site gives independent relaxation, while the common $\sqrt{\gamma}$ sets the rate. This is *uniform local relaxation*, the reference point from which we vary environmental organization.

2.3 Generator-level dissipator geometry

The reference process exposes two operational coordinates. A *jump family* changes which combinations of the system couple directly to the environment. A *rate profile* redistributes the available coupling weight within a fixed family.

The distinction between strength and organization is easiest to see with two qubits. Independent local relaxation uses two separate channels,

$$L_1 \propto \sigma_1^-, \qquad L_2 \propto \sigma_2^-,$$

so that the environment acts on the two sites individually. A shared relaxation channel instead uses one combined operator,

$$L_c \propto \sigma_1^- + \sigma_2^-.$$

Here the two relaxation amplitudes are combined before the environmental action occurs. To see the consequence, consider the one-excitation states

$$|+\rangle = \frac{|10\rangle + |01\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|10\rangle - |01\rangle}{\sqrt{2}}.$$

For the symmetric state $|+\rangle$, the two contributions add, and the shared channel directly connects the state to $|00\rangle$. For the antisymmetric state $|-\rangle$, they cancel, so this channel does not relax the state directly. The state is not necessarily protected forever: the Hamiltonian can mix the symmetric and antisymmetric combinations, thereby giving the latter an indirect route to the environment. Nevertheless, the two dissipative organizations are physically different. Independent channels expose each site separately, whereas a shared channel exposes only one collective combination directly.

Figure 1 maps these choices across the tested design space. The local–collective example is intuitive, but the distinction must be defined at the generator level.

Let $\{F_\mu\}$ be a fixed set of traceless system operators with

$$\mathrm{Tr}(F_\mu^\dagger F_\nu) = \delta_{\mu\nu}.$$

Expanding every jump operator as

$$L_k = \sum_\mu A_{k\mu} F_\mu, \qquad C = A^\dagger A \succeq 0,$$

makes this distinction invariant. The coefficient $A_{k\mu}$ states how strongly elementary action F_μ contributes to environmental channel k. The matrix C records which elementary actions are coupled and how the coupling weight is arranged among them. Equivalent remixings of the jump list change the individual L_k and A, but leave C, and therefore the dissipative generator, unchanged. These coordinates are operational rather than globally orthogonal: a family fixes the operator sector and allowed support of C, whereas within-family parameters set its nonzero weights and, where applicable, its direction within that support. The construction is invariant under equivalent jump-list remixings but remains relative to the declared elementary-operator basis.

For the one-body lowering block, the local–collective distinction takes the form

$$C_{\text{lower}}^{\text{local}} \propto I_N, \qquad C_{\text{lower}}^{\text{collective}} \propto \boldsymbol{c}\boldsymbol{c}^\dagger,$$

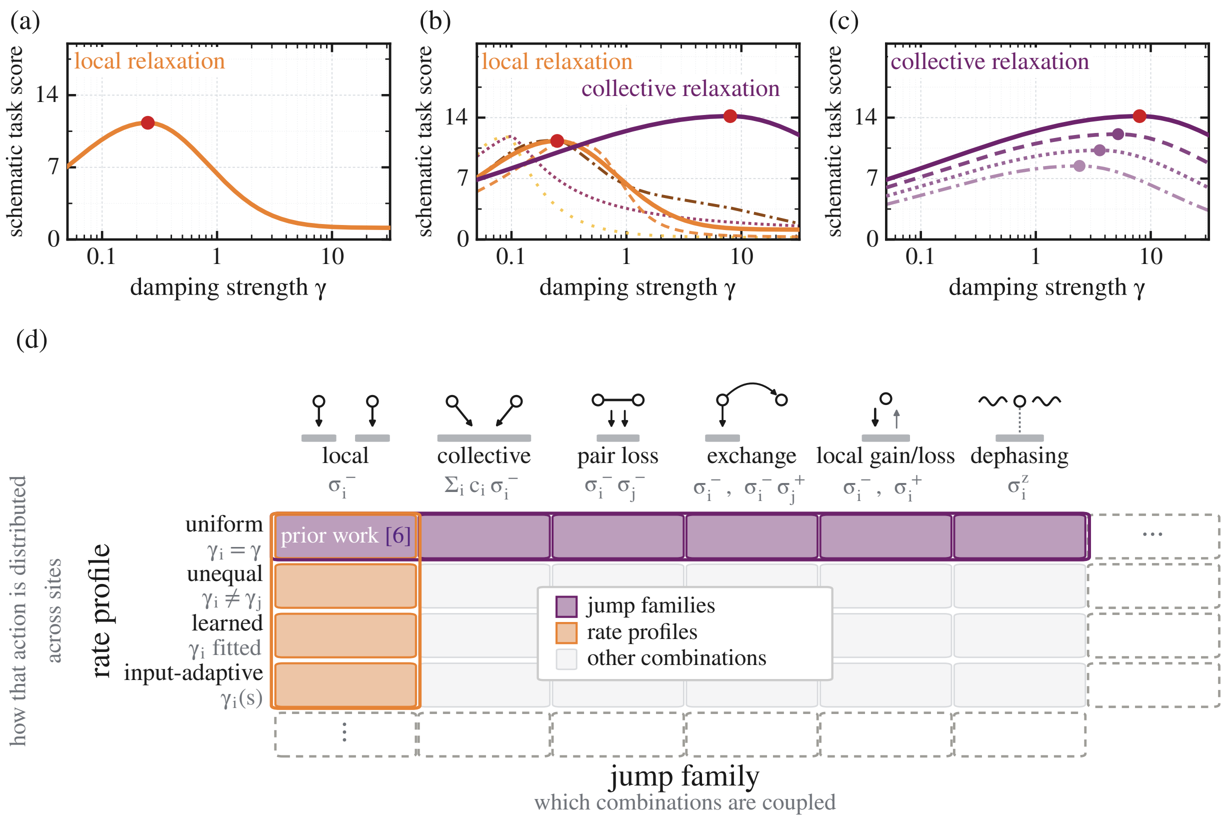

Figure 1: **Dissipator geometry separates two complementary design coordinates.** *(a)* Previous work primarily varies the overall strength of uniform local relaxation [6]. *(b)* The jump family changes which system combinations couple directly to the environment. *(c)* The rate profile redistributes coupling weight within a selected family. *(d)* Highlighted cells mark the two principal coordinate slices; pale-gray cells show other combinations. Panels (a–c) show schematic task scores, not fitted task data.

where $\boldsymbol{c} = (c_1, \ldots, c_N)^\mathsf{T}$ parameterizes the shared operator

$$L_c \propto \sum_i c_i \sigma_i^-.$$

With nonzero local rates, I_N spans N independently coupled lowering directions, whereas $\boldsymbol{c}\boldsymbol{c}^\dagger$ spans one directly coupled collective direction. Rotating the written local jump operators cannot turn this full-rank block into the rank-one shared block. Thus local and collective relaxation are distinct generators.

Unequal local relaxation instead changes the diagonal weights within the local block and thus moves along the rate-profile axis.

Table 1 is the exact catalogue of the tested environmental actions; Fig. 1 provides their intuitive map. We say gain/loss rather than thermalization because its fixed transitions need not produce a thermal state.

2.4 Common structural normalization and paired comparison

Unlike families contain different numbers and types of jump operators. Assigning the same coefficient to every operator would therefore give more total weight to a family merely because it contains more operators. We instead fix

$$\mathcal{B} = \sum_k \mathrm{Tr}(L_k^\dagger L_k). \tag{4}$$

Each term is the squared operator size assigned to one environmental channel. In the orthonormal basis above, $\mathcal{B} = \mathrm{Tr}\, C$. Fixing $\mathcal{B}$ therefore provides a common structural bookkeeping scale: the total assigned operator weight remains fixed while its organization changes.

For the principal fixed-$\mathcal{B}$, common-protocol comparison, we vary only the jump family. The Hamiltonian instance, input sequence, Pauli observables, linear readout, training, selection, and test splits, and assigned operator weight $\mathcal{B}$ remain fixed.

This isolates structural change, not physical equivalence. In the one-body local–collective comparison, fixed $\mathcal{B} = \operatorname{Tr} C$ holds assigned coupling weight fixed while changing the cross-site dissipative correlations encoded in C; it does not equalize activity, the relaxation spectrum, energy flow, entropy production, implementation, or hardware cost. The fixed-$\mathcal{B}$ protocol therefore defines the cross-family structural slice, while activity, gap, and operating-point controls additionally validate only the local–collective memory ordering.

2.5 Readout and computational tasks

After each input interval, local Pauli expectations and same-axis two-qubit correlations summarize the reservoir state,

$$\{\langle X_i\rangle, \langle Y_i\rangle, \langle Z_i\rangle, \langle X_iX_j\rangle, \langle Y_iY_j\rangle, \langle Z_iZ_j\rangle\}.$$

They provide $3N + 3\binom{N}{2}$ features, or 45 at $N = 5$. Fixed-profile structural experiments fit only the linear readout; separately identified profile and operating-point studies also select dissipative parameters on data disjoint from final testing. Appendix B gives the ridge control.

Four benchmarks ask what information those features retain. Short-term memory (STM) [24] reconstructs earlier inputs. NARMA-10 is the nonlinear autoregressive moving-average task of order ten. It tests a nonlinear function of recent input history [25]. Parity combines binary memory with nonlinear processing. Mackey–Glass forecasting [26] continues a chaotic signal in closed loop. Capacity is maximized for STM and parity. Error is minimized for NARMA-10 and forecasting.

For the adaptive-profile, finite-sampling, and alternative-control studies, any additional mean matching, estimator choice, calibration, or selection rule is stated explicitly.

Table 1: **Dissipative designs.** The first two rows differ only in rate profile. The displayed jump operators are convenient representatives of the corresponding GKLS generators. Equivalent jump decompositions are not counted as different designs.

design	operators and direct environmental action
uniform local relaxation [6]	$L_i = \sqrt{\gamma}\,\sigma_i^-$, $i = 1, \dots, N$. The same rate for every qubit; independent sitewise relaxation and the reference design.
unequal local relaxation	$L_i = \sqrt{\gamma_i}\,\sigma_i^-$, $i = 1, \dots, N$, with the γ_i not all equal. Independent sitewise relaxation with qubit-dependent timescales.
collective relaxation	$\{\sum_i c_i\sigma_i^-\}$. One shared relaxation channel.
pair loss	$\{\sigma_i^-\sigma_j^-\}_{i<j}$. Joint two-excitation removal.
exchange-assisted relaxation	$\{\sigma_i^-\} \cup \{\sigma_i^-\sigma_j^+\}_{i\neq j}$. Local lowering supplemented by incoherent excitation exchange.
local gain/loss	$\{\sigma_i^-, \sigma_i^+\}$. Fixed downward and upward transitions.
dephasing	$\{\sigma_i^z\}$. Phase randomization without excitation loss.

3 Computational consequences of the two-axis dissipator design

Section 2 defined dissipator geometry through the jump-family and rate-profile coordinates. We now traverse the design space of Fig. 1 in three steps: first across dissipator structures, then across rate profiles within a selected structure, and finally across both axes jointly. We subsequently ask whether the resulting exact-expectation differences remain visible under finite measurement budgets.

Rather than seeking one universally optimal dissipator, we test whether these two coordinates produce task-dependent specialization.

3.1 Reference model and how to read the results

Uniform local relaxation is the reference, with one equal-rate lowering channel per qubit. Its family also includes unequal, learned, and input-adaptive profiles. The main comparison uses the equal-phase rank-one collective channel with $c_i = 1$ (Appendix A); profile-optimized models instead learn within-family coefficients.

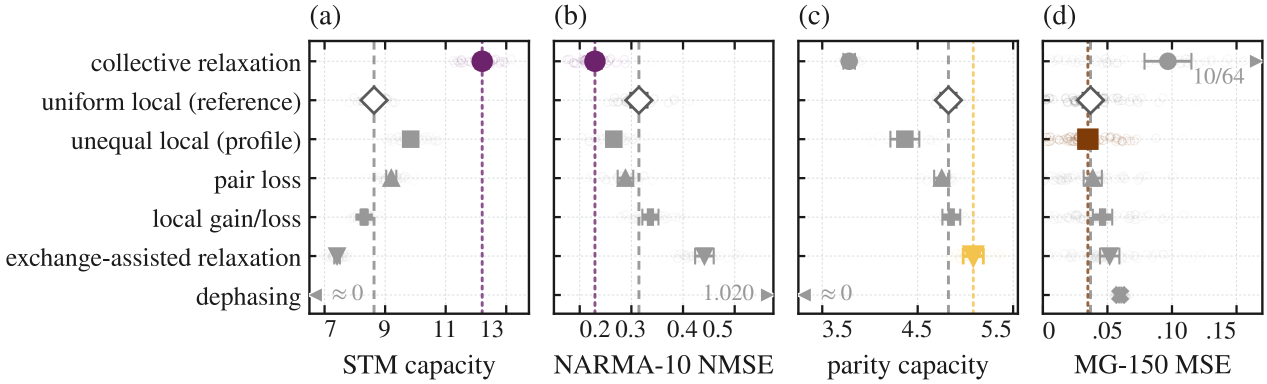

Figure 2: **Dissipator structure changes the computational task profile.** Absolute $N = 5$ scores for continuously driven designs. Rings show seeds; symbols and bars show means and pointwise 95% intervals. Dark-gray diamonds and dashed lines mark uniform local; saturated markers and matching dotted lines mark each best mean. Focused scales retain all non-dephasing seeds in (a–c), with edge labels for off-scale dephasing; in (d), 10/64 collective seeds exceed 0.17. Best-mean highlights are not significance tests; rankings are fixed-$\mathcal{B}$, common-point comparisons, not claims of physical equivalence. Unequal local is profile-only.

The four tasks retain their native metrics: STM and parity capacities are maximized, whereas NARMA-10 NMSE and MG-150 MSE are minimized.

Figure 2 reports the absolute $N = 5$ scores in each task's native metric and favorable direction. The dashed vertical line in each panel marks the uniform-local mean; the matching colored dotted line and saturated marker identify the best mean. Faint rings show individual reservoir instances. The panels retain their own scales and must be read independently; Appendix Table 2 gives the exact means and standard errors.

3.2 Axis I: Dissipator structure changes the task profile

We first ask whether changing which system combinations couple to the environment alters what the reservoir can compute.

Each paired comparison shares its Hamiltonian, input sequence, operating point, observables, readout, data split, and structural budget $\mathcal{B}$. Only the dissipator structure changes. Uniform local relaxation supplies the reference, while the reset-encoded Fujii–Nakajima (FN) architecture [3] is shown separately as an external reservoir baseline.

Appendix Figure 6 gives the complementary within-task rank view.

The resulting task profiles differ, and no continuously driven design dominates all four tasks. Collective relaxation leads STM and NARMA-10 under the common protocol, but performs worse on parity and long closed-loop forecasting. Exchange-assisted relaxation provides the clearest parity improvement.

Under fixed $\mathcal{B}$, the map shows protocol-specific specialization, not a universal winner. A descriptive STM strength sensitivity preserves the collective lead on the tested grid (Appendix B.8). We focus on the local–collective contrast because it is the cleanest comparison within one lowering action and gives the strongest recurring memory change.

3.3 Main structural case study: collective relaxation versus the reference model

We compare complete fixed-budget endpoints with $C^{\mathrm{local}}_{\mathrm{lower}} \propto I_N$ and $C^{\mathrm{collective}}_{\mathrm{lower}} \propto \boldsymbol{c}\boldsymbol{c}^{\dagger}$; rank, delocalization, coefficient-phase pattern, and drive alignment are not identified as separately causal.

At $N = 5$, collective relaxation increases STM from 8.634 to 12.206. This is an absolute paired gain of 3.57 STM-capacity units and a relative increase of 41%, with a 95% interval of $[3.36, 3.78]$ and 32/32 paired wins. NARMA-10 NMSE decreases from 0.315 to 0.229, corresponding to a 27% reduction in error, and improves in every paired reservoir.

Figure 3(b) provides a finite-size recurrence check of the principal $N = 5$ ordering, not an asymptotic scaling analysis. Across 24 paired lineages per size, collective relaxation has the highest mean STM and the lowest NARMA-10 error from $N = 4$ through $N = 8$.

Before interpreting the larger STM score as useful input memory, we test what is being remembered. Removing every jump operator while retaining the Hamiltonian, input protocol, and readout leaves STM and parity near zero, raises NARMA-10 NMSE above one, and preserves initialization dependence (Appendix Table 2). Coherent driving alone therefore reproduces neither the task profile nor controlled forgetting. This unmatched diagnostic contrasts with reset FN, which supplies non-unitary forgetting.

We next distinguish input memory from persistence of the starting state. The same switched input sequence is applied from four widely separated states, and their maximum trace distance is tracked. The convergence records in Appendix B show that local relaxation and pair loss converge across $N = 4, 5, 6$ and remain converged through the 1200-input continuation.

Collective relaxation converges more slowly, but reaches the same regime at the principal $N = 5$ setting and in the tested $N = 6$ lineages.

At $N = 5$, it approaches numerical precision by the strict 800-input washout and reaches numerical precision during the specified continuation to 1200 inputs.

A separate ten-pair $N = 5$ task run then uses the 800-input washout before scoring. Cross-initialization differences are negligible, while collective relaxation retains $+3.564\,[3.124, 4.004]$ STM units with 10/10 wins (Fig. 3(a)). After the tested initial states have converged, the additional retention therefore belongs to the input history rather than to the starting state.

The comparison is also repeated under four spin-Hamiltonian ensembles that change the interaction axis, drive axis, interaction form, or connectivity. Every ensemble retains a positive collective-minus-reference STM difference, ranging from +2.22 to +4.02 units (Fig. 3(a)). These replications preserve the continuous encoding and Pauli readout, so they establish recurrence within the tested driven-spin architecture. Appendix B describes the Hamiltonian variants and ridge control.

To test whether the ordering is tied to continuous-drive encoding, we replace it with input-by-reset while retaining the $N = 5$ $XX + Z$ processor, Pauli readout, task definitions, and fixed structural budget. After an 800-input washout, collective relaxation increases STM from 4.596 to 6.369, a gain of $1.773\,[1.312, 2.235]$, and decreases NARMA-10 NMSE from 0.673 to 0.491, a favorable reduction of $0.182\,[0.144, 0.220]$, with 16/16 favorable pairs for both endpoints (Fig. 7 and Appendix B.3). The local–collective ordering therefore recurs across two tested input architectures rather than being an artifact of continuous driving.

3.4 Axis II and joint design: rate profiles refine the chosen structure

Choosing a dissipator structure does not fully specify the environmental coupling. A second choice remains: how should the available coupling weight be distributed within that structure? We first keep the local-relaxation family fixed and compare uniform, random unequal, learned static, and input-adaptive rate profiles. All sites relax in every candidate, but they need not do so equally. The profile-only unequal-local row in Appendix Table 2 reports the common-protocol comparison; the dedicated STM sweep below uses the separately specified profile-optimization protocol.

The uniform profile assigns the same rate to every site. A random unequal profile introduces a fixed distribution of local timescales. A learned static profile is fitted separately to each reservoir using validation data. An adaptive profile varies with the current input. The static profiles share the same mean rate. The adaptive profile is mean-matched over a fixed input grid but may vary with the input.

Figure 4(a) shows that rate redistribution changes STM without changing the local-relaxation family. Random unequal rates improve on the uniform-local reference, and reservoir-specific learning improves further. The learned-minus-uniform gain is 1.34 units, with an interval of $[1.13, 1.55]$ and 32/32 wins. The adaptive profile gives only a marginal additional gain under the matched restart and evaluation limits. This ordering does not establish a general superiority of static profiles because the static and adaptive parameterizations differ in dimension and stopping behavior.

Within-family tuning redistributes diagonal local rates but changes the supported rank-one collective direction; both remain reproducible tuning choices.

The structure and profile axes are not competing alternatives. We therefore cross them directly

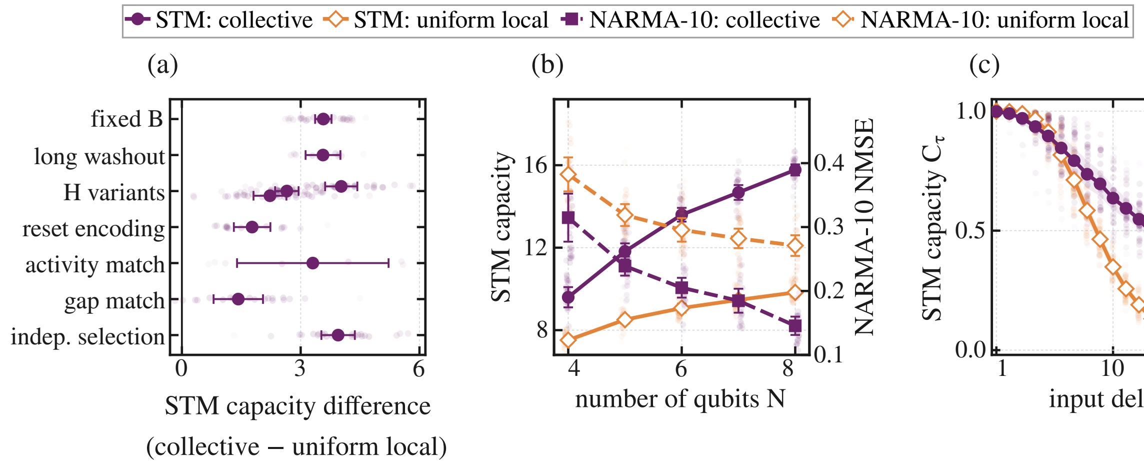

Figure 3: **Collective relaxation retains more usable recent-input memory than the uniform-local reference.** *(a)* Collective-minus-uniform-local STM-capacity differences under targeted controls; the three alternative-Hamiltonian estimates remain separate. *(b)* Fixed-initialization size sweep for $N = 4, \dots, 8$. *(c)* Delay-resolved STM capacities. Faint marks and lines show paired effects in (a) and seed-level absolute scores or capacities in (b,c); bars and bands give the declared 95% intervals.

by comparing uniform and learned profiles within both local and collective relaxation at the same $\mathcal{B}$. Figure 4(b) displays this 2×2 comparison. The vertical separation shows the structural effect, the uniform-to-learned slope shows the profile effect, and the different slopes show that the profile benefit depends on the selected structure.

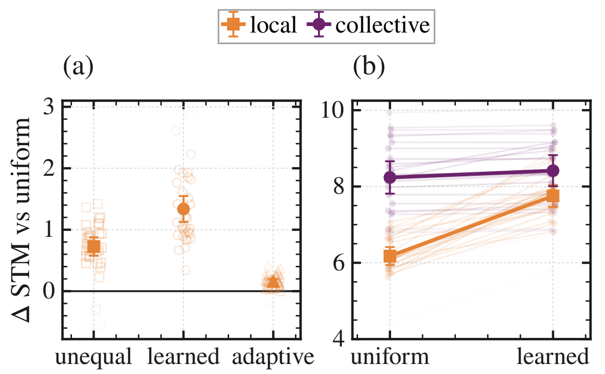

Figure 4: **Rate profiles refine the selected dissipator structure.** *(a)* All 32 paired STM effects for three local rate profiles. Large markers and bars give the mean and 95% interval. *(b)* Paired uniform-to-learned trajectories for local and collective relaxation, showing family-dependent profile gains. Seed marks and trajectories become lighter with distance from their condition means.

The profile gain is larger for local relaxation. The difference between the two gains is 1.397 STM units; all 24/24 paired values are positive, and the familywise 95% interval is $[1.078, 1.716]$. Collective relaxation nevertheless remains higher after both profiles are learned. Profile optimization refines the selected structure by a family-dependent amount. Appendix C gives the parameterizations and inference scope.

3.5 Finite sampling determines experimental visibility

The preceding comparisons characterize the computation available to an exact 45-Pauli readout. An experiment observes finite measurement outcomes rather than exact expectation values. We therefore ask a distinct question: at what preparation budget do the differences created by dissipator geometry become statistically visible?

Finite-sample training, resolvable expressive capacity, and statistical-noise robustness have been studied in QRC and physical learning systems [27, 28]. Our measurement study asks whether the observed ordering survives when expectation values must be estimated from the same nominal number of state preparations. It does not model a platform-specific implementation cost.

At each step, independent sampling spends $N_{\text{prep}} = 45q$ shots over 45 observables; grouped sampling spends the same nominal budget in the global X, Y, and Z bases and reuses each bitstring across 15 features. Each design validates its ridge coefficient before one held-out evaluation. "Preparations per step" excludes replay of the preceding input history, so Fig. 5 studies measurement resolution, not end-to-end runtime.

At the exact-expectation endpoint, Fig. 5 recovers the collective-over-reference memory ordering from the structural comparison. Grouped sampling resolves this contrast with one-sixteenth of the preparation count needed by independent sampling. This gain is close to the 15-fold feature reuse built into the estimator. It is a measurement advantage rather than a hardware advantage specific to collective relaxation.

The finite-data curves also show why measurement design belongs in the comparison. Sampling noise does not reduce every STM score by the same amount. At the smallest independent-sampling budgets, pair loss appears best. Unequal local relaxation leads at intermediate precision. Collective relaxation emerges only after the observables are estimated accurately enough. Under simultaneous inference across the complete comparison family, grouped sampling resolves collective relaxation above every competitor only at the largest finite budget. Independent sampling never resolves it as best on the tested grid.

Appendix D gives the estimator definitions, budget parameterization, and simultaneous-inference scope.

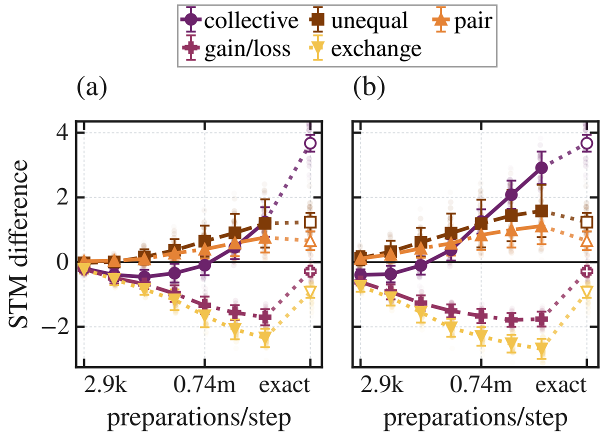

Figure 5: **Finite sampling determines when the dissipator-structure ordering becomes visible.** *(a,b)* Paired STM differences from uniform local relaxation at equal preparation budgets under independent and grouped sampling, respectively. Open markers denote exact readout. Grouping recovers the exact ordering sooner by reusing compatible observables. Faint points show individual paired-seed differences and become lighter with distance from the corresponding mean.

Across these analyses, the memory advantage of collective over uniform local relaxation is the strongest recurring structural effect. Section 4 now examines which dynamics are consistent with that contrast.

4 Dynamical interpretation of the local–collective memory contrast

Section 3 establishes the controlled computational ordering independently of the interpretation below. Here we ask which dynamical picture is consistent with it after the tested initial states have been forgotten, not which microscopic ingredient is uniquely causal.

4.1 Working physical picture

The simplest distinction concerns how the reservoir reaches its environment. Under uniform local relaxation, every spin has its own direct relaxation channel. Under collective relaxation, the spins share one channel. The same assigned coupling weight is therefore spread over several separate paths in one case and concentrated into a common path in the other. Some combinations of spin excitations then reach the collective channel only after the Hamiltonian has mixed them with the directly coupled combination. They are not permanently protected: coherent dynamics continue to move information between these combinations. Appendix A gives the corresponding mathematical formulation and states its precise limits. The same interference principle appears in collective emission and in network states that a shared environment cannot address directly [19, 20].

This gives the working picture. A shared environmental channel directly couples fewer combinations; Hamiltonian mixing provides indirect relaxation routes for the others; the resulting slower readout-visible responses can preserve a longer accessible input history.

4.2 Scalar controls for the memory contrast

Fixing $\mathcal{B}$ controls assigned operator weight, but it does not equalize realized jump activity, the relaxation spectrum, or the operating point preferred by each architecture. We therefore ask whether any one of these scalar differences is sufficient to explain the principal memory contrast. These controls are not intended to factorize the generator into uniquely causal structural ingredients. Figure 3(a) summarizes the fixed-$\mathcal{B}$, activity-matched, gap-matched, and independently selected comparisons. The STM contrast is positive in all four protocols.

The first control asks whether the shared channel wins simply because it produces a different amount of realized jump activity. Local and collective rates are calibrated separately to the same time-averaged expected activity on an input stream without task labels, then frozen before task scoring. Every calibration passes and the held-out activity difference is unresolved. Collective relaxation still gains 3.306 STM units, with a 95% interval of $[1.393, 5.220]$ and 8/8 paired wins. Different mean jump activity cannot by itself explain the memory contrast. Appendix B.5 gives the calibration protocol.

The second control asks whether the shared channel remembers longer simply because it relaxes more slowly. Before any task stream is generated, each local rate is therefore calibrated to the collective generator's Liouvillian gap at the constant input $s = 0.5$. This gap is the slowest nonzero decay rate of that constant-input evolution and provides one measure of relaxation speed. All 24 matches fall within the prescribed tolerance.

Gap matching reduces the STM and NARMA-10 contrasts by 62.1% and 48.2%, respectively, showing that this timescale explains an important part of the effect. It does not remove the ordering. Collective relaxation retains 1.424 STM units with an interval of $[0.800, 2.049]$ and 20/24 wins, and the NARMA-10 difference remains resolved. Appendix Figure 8(b) resolves the same comparison by input delay. Matching the local rate restores much of its memory tail, but collective relaxation remains higher at long delays. The bands are pointwise 95% intervals across the 24 paired reservoirs. Appendix B.6 gives the calibration protocol.

Finally, a common setting could accidentally favor one jump family. Local and collective relaxation therefore choose their own drive, input duration, dissipative multiplier, and ridge coefficient, with disjoint data for selection and final testing. Collective relaxation still leads by 3.945 STM units, with an interval of $[3.523, 4.368]$ and 24/24 wins. The contrast does not require a shared operating point. The search is bounded rather than global; Appendix B.7 gives the selection protocol and bounded-search scope.

These controls rule out unequal mean activity, one representative decay rate, and a common operating point as sole explanations of the principal local–collective ordering. They do not equalize the complete spectrum, input-dependent propagator products, non-normal transients, energy flow, entropy production, implementation effort, or hardware cost, and they were not repeated for the full cross-family task map.

4.3 Positive dynamical support

Two established ideas guide this interpretation. Fading memory asks whether old inputs and the initial state gradually lose their influence [2, 29, 30]. Information-processing capacity asks which functions of recent input history remain available to a readout [31, 32]. Quantum versions of these ideas have been developed for finite, input-driven, and subsystem reservoirs [33, 34, 35]. Relaxation spectra provide a complementary fixed-input view of how an open quantum system forgets [36, 37]. Because switched inputs mix those responses, we use the spectral calculations as diagnostics rather than as a complete mechanism.

Because STM is obtained by linear regression on the measured feature trajectory, a dynamical component can contribute only if the input excites it and it remains visible in the readout span. Fixed-input mode overlap tests readout visibility; the switched-response kernel additionally includes input excitation. The slower collective modes remain Pauli-visible, and after the representative timescale is matched the switched response is concentrated into fewer observable patterns.

Two additional checks connect this response to dissipator geometry. Across six process designs, slower midpoint relaxation usually accompanies more STM. A separate sweep mixes local and collective loss continuously; its selected interior point improves on local relaxation in every fresh pair, demonstrating robustness to residual local damping along this path. Together, the diagnostics support one coherent account of the endpoint contrast; Appendix E reports their uncertainty.

4.4 Scope of the explanation

Along the fixed-weight interpolation, every interior one-body lowering block retains Kossakowski rank N while STM changes, so rank alone is insufficient. Learned profiles and the interpolation change several coordinates together. Two matched rank-one interventions isolate direction more sharply. The primary $N = 5$ study finds more STM along the equal-phase end of a prespecified path and against five zero-overlap controls. In a separately generated $N = 6$ replication, rotating the equal-phase vector to the real, sign-balanced direction $(1, 1, 1, -1, -1, -1)$, orthogonal to the uniform site profile of the input drive, lowers STM by 2.271 units $[1.826, 2.717]$, with 24/24 paired wins for equal phase and exact two-sided sign-test $p = 1.19 \times 10^{-7}$. Kossakowski rank, nonzero spectrum and trace, assigned operator weight, sitewise diagonal weights, coefficient magnitudes, and every within-pair processor and data choice remain fixed (Appendix F). These invariants are therefore insufficient to determine memory: dissipative orientation relative to the fixed driven processor is an operative architectural coordinate. This does not distinguish alignment with the input from alignment with the coherent dynamics, or make equal phase or a response diagnostic universal.

Dephasing follows a different contraction mechanism. In the connected primary model, it removes the measured signal after washout, which accounts for its near-zero STM. Appendix E gives the driven-dephasing argument. It should not be folded into the shared-channel explanation.

5 Discussion, scope, and conclusion

In the tested driven-spin reservoir, environmental coupling cannot be summarized by one scalar: its organization changes which traces of earlier inputs remain accessible to the readout. Dissipator geometry is therefore an architectural degree of freedom.

Under input-by-reset encoding, NARMA-10 after the 800-input washout favors collective relaxation in all 16 fresh pairs; the separate continuously driven long-washout STM control favors it in all ten. The STM advantage also survives activity and gap matching and bounded operating-point selection with disjoint selection and final-test reservoirs. Thus the local–collective advantage is not confined to the standard washout, one tested encoding, or one common operating point, although these controls do not cover every jump family. The ordering also recurs across four Hamiltonian ensembles and persists after the tested initial-state dependence has vanished. The diagnostics and tested local-to-collective interpolation support—but do not uniquely establish—the shared-channel account; the selected interior point retains the memory advantage despite residual local loss. Matched rank-one direction interventions at two finite sizes further show that rotating only the supported collective direction reduces STM relative to the equal-phase channel.

Within the common fixed-$\mathcal{B}$ slice, the tested jump families produce distinct task profiles rather than a globally optimized ranking. Appendix B.8 reports the complementary six-design STM strength sensitivity. Rate redistribution changes STM within a family, and finite-sampling precision determines which exact-expectation contrasts are statistically resolvable.

These claims remain bounded by the tested protocols. The full task map is an $N = 5$ driven-spin comparison; the Hamiltonian variants retain continuous encoding and the Pauli readout, while the reset test changes encoding but retains the $N = 5$ $XX + Z$ processor. Fixed $\mathcal{B}$ controls assigned operator weight rather than full dynamical or physical equivalence. The sampling study includes projection noise and estimator design, but not latency, drift, gate errors, or readout errors.

The appendix proves uniform contraction for driven dephasing, not the local–collective effect. The latter recurs in the fixed-initialization sweep across 24 fresh pairs at every tested size from $N = 4$ to 8, with higher mean STM and lower mean NARMA-10 error under collective relaxation. The $N = 4, \dots, 8$ sweep supports recurrence across the tested exact finite reservoirs; it is not used to infer a scaling exponent or persistence in the thermodynamic limit. The $N = 4$ boundary case is excluded from initialization-independent claims, and the sweep is not an analytical proof of the collective mechanism.

This design view connects QRC to the broader use of engineered environments for state preparation, quantum information, and memory protection [38, 39, 40, 41]. Shared resonators, waveguides, and structured couplers motivate correlated-dissipation engineering [42, 43, 44, 45], but an arbitrary common bath need not realize the exact bare-spin jump used here. The central advance is an architecture-level computational result: even after matching Kossakowski rank, nonzero spectrum and trace, assigned operator weight, diagonal weights, coefficient magnitudes, and the processor protocol, rotating the supported dissipative direction changes readout-accessible memory.

Controlled forgetting is therefore a design choice for the memory and computation made accessible by a quantum reservoir.

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A Reservoir model, dissipator construction, and shared methods

This appendix gives the shared definitions and numerical conventions, then the additional controls and interpretation behind each analysis.

A.1 Data and code availability

All code, data, and analysis workflows for this work, together with the manuscript source and figure inputs, are available at `https://github.com/eybmits/qrc-dissipation-engineering`. The repository documents the full protocol, including random seeds, Hamiltonian and input definitions, data splits, parameter grids, solver and optimization settings, and numerical tolerances. It also contains the calibration records, convergence checks, and per-run outputs behind each reported number, so every result in the paper can be inspected and reproduced.

A.2 Collective process, pairing, and readout

The principal collective process uses one fixed vector in the main, finite-size, Hamiltonian, activity-matched, and gap-matched comparisons:

$$c_i = 1 \quad (i = 1, \ldots, N), \qquad \boldsymbol{c} = (1, \ldots, 1), \qquad L_{\mathrm{c}} = \sqrt{\gamma} \sum_{i=1}^{N} \sigma_i^- .$$

Thus the coefficients are real, have equal phase, and satisfy $\sum_i |c_i|^2 = N$. Only the separately identified profile-by-family check fits a nonuniform collective profile. The common target is the unit-rate local value $\mathcal{B} = N2^{N-1}$, giving $\mathcal{B} = 80$ at $N = 5$. The finite-size study rematches this target separately at every N.

The jump-family comparison uses independently drawn complete-graph couplings $J_{ij} \sim U[-1, 1]$. Unless a control explicitly changes an operating point, $h = \Delta t = 0.5$, the reservoir starts in $|0 \cdots 0\rangle$, and the STM and NARMA-10 sequences contain 200 washout, 600 training, and 400 test inputs. The task-specific parity and MG-150 protocols are stated below. Comparisons are paired: for a given seed, all dissipative designs see the same Hamiltonian, input sequence, targets, and data split. Variation across seeds therefore reflects both a new reservoir and a new input sequence.

At $N = 5$, the readout contains the expectations of all one-qubit Pauli operators and all same-axis two-qubit Pauli products. These 45 measured features, together with a bias, feed a linear classical readout. The internal quantum dynamics are not trained. Continuous-input evolution is computed by sparse matrix exponentiation. The dense-grid approximation used only for the bounded local-profile search is described in Appendix C.

A.3 Tasks and what they probe

Short-term memory (STM) asks whether a past input can be reconstructed from the present reservoir state. We sum the squared-correlation capacities for delays $1, \ldots, 20$. Larger values mean that more of the recent input history remains linearly accessible.

NARMA-10 adds nonlinear temporal processing. Its target obeys

$$\begin{aligned} y_k = 0.3 y_{k-1} + 0.05 y_{k-1} \sum_{j=1}^{10} y_{k-j} \\ + 1.5 \tilde{s}_{k-10} \tilde{s}_{k-1} + 0.1, \qquad \tilde{s} = 0.2 s, \end{aligned}$$

and is scored by normalized mean-squared error (NMSE), so smaller is better. Parity uses binary input windows of length $1, \ldots, 7$ and 15 virtual nodes per input interval. It has separate 500/250/400 train/validation/test streams after 100 washout inputs. Validation selects the readout regularization before a train-plus-validation refit.

Table 2: **Absolute scores for the fixed-$\mathcal{B}$, common-protocol comparison.** Entries are mean ± standard error. Arrows give the favorable direction, and bold marks the best mean. Row groups distinguish external or diagnostic comparisons, fixed-profile structural models, and a profile-only variation within local relaxation. Task definitions and sample sizes are stated immediately above.

model	STM ↑	NARMA-10 ↓	parity ↑	MG-150 ↓
External and diagnostic comparisons				
no dissipation	0.047 ± 0.004	1.308 ± 0.036	0.016 ± 0.001	0.0697 ± 0.0016
reset-encoded FN [3]	10.106 ± 0.260	0.367 ± 0.012	2.295 ± 0.125	0.115 ± 0.008
Fixed-profile structural models				
uniform local relaxation [6]	8.634 ± 0.069	0.315 ± 0.008	4.822 ± 0.038	0.0369 ± 0.0029
collective relaxation	$\mathbf{12.206 \pm 0.101}$	$\mathbf{0.229 \pm 0.006}$	3.782 ± 0.028	0.0969 ± 0.0091
pair loss	9.203 ± 0.082	0.289 ± 0.007	4.752 ± 0.037	0.0387 ± 0.0035
local gain/loss	8.307 ± 0.058	0.337 ± 0.008	4.852 ± 0.044	0.0462 ± 0.0038
exchange-assisted relaxation	7.413 ± 0.045	0.441 ± 0.009	$\mathbf{5.083 \pm 0.049}$	0.0518 ± 0.0038
dephasing	0.000 ± 0.000	1.020 ± 0.006	0.000 ± 0.000	0.0599 ± 0.0004
Profile-only variation within local relaxation				
unequal local relaxation	9.845 ± 0.119	0.266 ± 0.007	4.366 ± 0.071	$\mathbf{0.0349 \pm 0.0028}$

The forecasting target obeys the Mackey–Glass equation

$$\dot{x}(t) = \frac{0.2x(t-17)}{1+x(t-17)^{10}} - 0.1x(t).$$

It is integrated by Euler steps of length one from the seeded delay history $x_t = 1.2 + 0.01\xi_t$, with $\xi_t \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$. Every third Euler iterate is retained, and the first 500 retained samples are discarded. After a 200-sample washout and 600-sample teacher-forced training prefix, the readout is closed for 150 steps. The affine scale is fixed by the washout-plus-training prefix. Only closed-loop predictions fed back to the reservoir are clipped to $[0,1]$, while the held-out target remains unclipped.

For Table 2, each continuously driven design and the external reset-FN baseline uses 32 reservoirs for STM and NARMA-10, 16 for parity, and 64 for MG-150. The no-dissipation diagnostic uses 30, 30, 32, and 32 reservoirs, respectively.

A.4 Constructing and normalizing dissipative processes

Table 1 gives the physical action of the six process families and the unequal-rate variant. Unequal local rates are drawn log-uniformly from $[0.1, 10]$ and rescaled to unit mean. The collective coefficients are the fixed equal-phase choice $c_i = 1$ stated in Appendix A.2. Pair loss contains every $i < j$. Exchange-assisted relaxation combines local lowering with ordered, excitation-conserving exchange jumps, assigning 40% of $\mathcal{B}$ to the former and 60% to the latter. The local gain/loss process uses downward/upward rates 1/0.3 before the common rescaling. After construction, one scalar rescales every completed design to the same $\mathcal{B} = \sum_k \operatorname{Tr}(L_k^\dagger L_k)$ as unit-rate uniform local relaxation.

Section 2 defines the fixed operator basis and Kossakowski matrix used to compare generators independently of jump-list remixing. For the one-body lowering families, write

$$L_k = \sum_i A_{ki}\sigma_i^-, \qquad \Gamma = A^\dagger A \succeq 0.$$

Then

$$\mathcal{B} = 2^{N-1}\operatorname{Tr}\Gamma.$$

Thus fixed $\mathcal{B}$ fixes $\operatorname{Tr}\Gamma$. Uniform local relaxation has rank N, whereas the equal-phase collective channel has rank one. Kernel directions are Kossakowski coupling directions, not decay modes of the full Liouvillian; Hamiltonian mixing can provide indirect relaxation. The identity above applies only to the one-body lowering block. Across unlike operator sectors, fixed $\mathcal{B}$ remains a structural control rather than an equivalence of spectra, flows, or implementation cost.

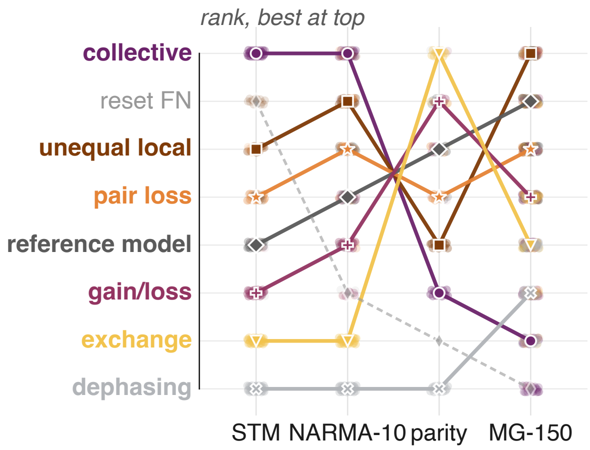

Figure 6: **Task-wise rankings confirm specialization rather than a universal winner among the tested continuously driven designs.** Ranks at $N = 5$ are induced by the absolute means in Table 2; rank 1 is best within each task. Uniform local is the reference. Colors follow the manuscript-wide dissipator palette; lines connect identical designs only as visual guides. Dashed reset-based FN is an external, unpaired baseline. Faint points rank the paired continuously driven seeds against the fixed reset-FN mean; no seed points are assigned to that baseline.

A.5 Statistical reporting

All primary effects are computed as paired, instance-level differences and oriented so that positive means improvement. Unless stated otherwise, we report the mean difference, a two-sided 95% Student-t interval, and the paired win count. For the non-reference cell comparisons, paired sign-flip tests are corrected over the comparison family with the Holm procedure [46]. Finite-size intervals use paired bootstrap resampling. The profile and finite-sampling experiments use simultaneous intervals because many related contrasts are examined together.

B Jump-family computation and local–collective controls

The jump-family analysis asks whether changing the jump family changes computation when the Hamiltonian, input, operating point, readout, and $\mathcal{B}$ are held fixed. The broad map establishes the task dependence of that structural choice. The subsequent checks focus on the principal collective-versus-local memory contrast.

B.1 Fixed-$\mathcal{B}$ structural task map

Figure 2 and Table 2 report the absolute fixed-$\mathcal{B}$ scores; Figure 6 gives the complementary rank view.

A validation-selected readout control reuses the same STM and NARMA-10 trajectories but divides the 600 fitting rows into 450 training and 150 validation rows. Each design and instance selects ordinary least squares or a ridge value before refitting on all 600 rows. The ordering is unchanged: the collective design remains best on both tasks. Thus the main ranking is not an artifact of imposing one readout penalty on every process.

B.2 Robustness to the coherent dynamics

The principal contrast was repeated in four paired $N = 5$ spin-Hamiltonian ensembles: the primary $XX + Z$ model with an X drive, a $ZZ + X$ model with a Z drive, an $XY + Z$ model with an X drive, and an XX ring with an X drive. The encoding and Pauli readout remain fixed.

B.3 Replication under reset-based input encoding

To test whether the local–collective ordering depends on continuous-drive encoding, we repeat the comparison in a reset-encoded reservoir. At input step t, qubit zero is replaced by

$$\rho_t^+ = |\psi(s_t)\rangle\langle\psi(s_t)|_0 \otimes \mathrm{Tr}_0(\rho_t), \qquad |\psi(s)\rangle = \sqrt{1-s}|0\rangle + \sqrt{s}|1\rangle,$$

after which the autonomous $XX + Z$ Hamiltonian evolves together with either uniform local relaxation or the equal-phase collective channel. This changes the input architecture: the reset is non-unitary and the input-dependent transverse drive is absent during the evolution interval.

The strict comparison otherwise retains the principal $N = 5$ construction: complete-graph couplings independently drawn from $U[-1, 1]$, $h = \Delta t = 0.5$, $\gamma = 1$, $\mathcal{B} = 80$, i.i.d. inputs $s_t \sim U[0, 1]$, the 45 Pauli features and bias, ridge 10^{-8}, STM delays $1, \ldots, 20$, the same NARMA-10 target, and 600 training and 400 untouched test inputs. Within each of 16 fresh pairs, the local and collective reservoirs share the Hamiltonian, input sequence, target, and data split. We use an 800-input washout because collective reset dynamics converge more slowly than their local counterpart. Table 3 reports the strict task endpoints, and Figure 7 shows their paired effects and lag-resolved STM profile.

Table 3: **Strict reset-encoded replication.** The effect is collective-minus-local for STM and local-minus-collective for NARMA-10, so positive values are favorable in both rows. Intervals are paired 95% confidence intervals.

task	local	collective	favorable effect	wins
STM capacity	4.595912	6.369394	1.773 [1.312, 2.235]	16/16
NARMA-10 NMSE	0.672973	0.490933	0.182 [0.144, 0.220]	16/16

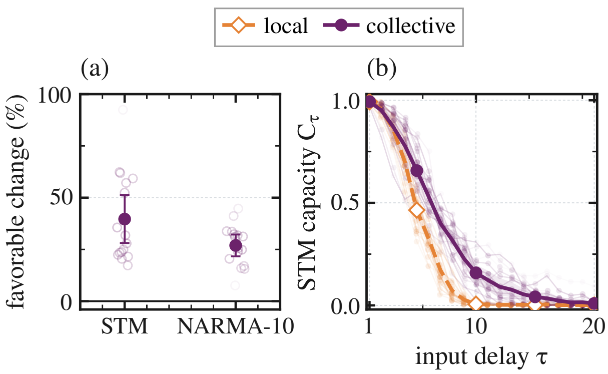

Figure 7: **Replication under reset-based input encoding.** *(a)* Favorable paired relative change with respect to local relaxation for STM capacity and NARMA-10 NMSE across 16 fresh paired reservoirs; positive percentages denote larger STM or lower NMSE. Points show seeds, and filled markers with bars show paired means and pointwise 95% t intervals. *(b)* Lag-resolved STM capacities after the strict 800-input washout; faint lines and points are seeds, and thick lines and bands show means and pointwise 95% t intervals. The fixed-budget comparison uses $\gamma = 1$ and $\mathcal{B} = 80$.

Repeating both tasks and both dissipative models from the ground, fully excited, maximally mixed, and Haar-random pure states gives a maximum score range below 7×10^{-7}; the largest trace distance at the 800-input washout is 1.3×10^{-14}, at numerical precision. The strict ordering therefore reflects retained input history rather than the tested starting state. It establishes that the effect transfers across the two tested input architectures, not universal architecture independence. Nor does this comparison imply that dissipation outperforms a reset-only unitary reservoir; it isolates how two matched dissipative organizations shape usable memory.

B.4 Finite size and loss of the initial state

The finite-size study uses 24 fresh paired lineages at every $N = 4, \dots, 8$. Nested principal submatrices of an $N = 8$ coupling draw are rescaled by $\sqrt{4/(N-1)}$, and the Frobenius reference is rematched within each size. The number of measured Pauli features, excluding the fitted bias, grows as

$$F(N) = 3N + 3\binom{N}{2}.$$

The sweep is intended as a finite-size robustness test of the paired local–collective ordering, not as an analysis of asymptotic memory, mixing-time, or Liouvillian scaling. STM is summed over a fixed 20-delay horizon and therefore satisfies $C_{\mathrm{STM}} \leq 20$ for every N, whereas $F(N)$ grows with N. Consequently,

$$\frac{C_{\mathrm{STM}}}{F(N)} \leq \frac{20}{3N + 3\binom{N}{2}}.$$

Even perfect recall at all tested delays would therefore have a decreasing STM-per-feature ceiling. This ratio is not a neutral scaling or resource-efficiency metric under the present protocol. The grouped readout still requires only the three global X, Y, and Z settings, while the fitted linear readout has $F(N) + 1$ inputs after including the bias. Figure 3(b) shows the absolute scores across size.

Because collective coupling can leave some combinations only indirectly damped, we separately tested whether the reported memory could be residual dependence on the starting state. For each process and size, eight fresh lineages at $N = 4, 5, 6$ were driven by identical switched input sequences from the ground, fully excited, maximally mixed, and a Haar-random pure state. Every checkpoint stores the largest trace and Pauli-feature distances over the six initial-state pairs. Local and pair processes converge rapidly. A continuation follows all 48 local and pair lineages at $N = 4, 5, 6$ to 1200 inputs; their worst trace distance over steps 1100–1200 is 6.85×10^{-15}.

For collective relaxation at the principal $N = 5$ setting, the worst trace distance falls from 4.78×10^{-4} after 200 inputs to 1.32×10^{-11} after 800. A separate continuation follows all eight lineages to 1200 inputs and reaches 4.42×10^{-15}, below its declared 10^{-14} gate. The slowly forgetting collective $N = 4$ boundary case is excluded from initialization-independent claims.

A stricter task-level check then uses ten fresh $N = 5$ reservoir pairs, three distant initial states, and washouts of 50, 200, and 800 inputs.

B.5 Matched expected jump activity

The fixed-$\mathcal{B}$ comparison deliberately allows the realized jump activity to differ. To test whether this alone explains the memory result, we calibrate the two processes using the time-averaged expectation

$$\mathcal{J} = \frac{1}{T\Delta t} \sum_{t=1}^{T} \int_0^{\Delta t} \sum_k \mathrm{Tr}\left[L_k^\dagger L_k \rho_t(\tau)\right] \, \mathrm{d}\tau.$$

Local and collective rates are fitted separately to $\mathcal{J}_\star = 0.5075$ on a shared, unlabeled input stream that is independent of every task stream. After 100 washout inputs, activity is averaged over 150 intervals. All 16 calibrations for eight fresh reservoir pairs pass the predeclared 1.5% tolerance before task scoring.

B.6 Matched midpoint Liouvillian gap

A second control asks whether the result is only a consequence of the slowest relaxation time at a representative constant input. For each of 24 fresh $N = 5$ Hamiltonians, the local rate is calibrated to the collective Liouvillian gap at $s = 0.5$ before task inputs are generated. All calibrations satisfy the 0.5% relative-error gate.

B.7 Independent operating-point selection

The third control lets local and collective relaxation select their own operating points without using test data. The bounded search varies the drive, input duration, dissipative strength, and readout penalty over predeclared grids. A two-reservoir prescreen advances eight settings per jump family. Twelve disjoint reservoirs then select one setting and readout penalty per process. The final 24 reservoirs are opened only after these choices are written. The collective strength choice is additionally bracketed on the selection set before the final test.

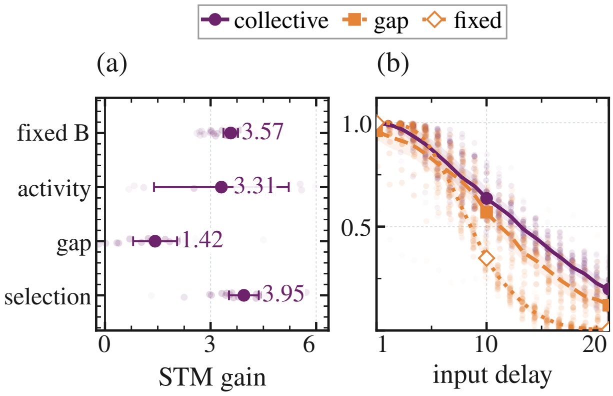

Figure 8: **Targeted scalar controls preserve the local–collective memory contrast.** *(a)* Collective-minus-reference STM effects for the fixed-$\mathcal{B}$, activity-matched, gap-matched, and independently selected protocols. *(b)* Lag-resolved STM at fixed $\mathcal{B}$ and after matching the local rate to the collective midpoint Liouvillian gap. Faint points show paired seeds; bars and bands show the declared 95% intervals.

B.8 Validation-selected scalar-strength sensitivity

The fixed-$\mathcal{B}$ family map is a controlled structural slice rather than a globally optimized ranking. In a complementary post hoc STM sensitivity analysis, scalar strength is selected separately for six non-dephasing process designs by leave-one-reservoir-out validation. All designs use the base multiplier grid $\{0.05, 0.1, 0.25, 0.5, 1, 2, 4\}$; uniform local and collective relaxation also include 8 and 16, which bracket the collective optimum at 8. The selected multipliers are 8 for collective, 0.1 for local gain/loss and exchange-assisted relaxation, and 0.25 for uniform local, unequal local, and pair loss; every curve is bracketed. Collective relaxation has the largest mean across the 20 held-out scores, and 50,000 selection-aware bootstrap draws give Bonferroni-simultaneous 95% percentile intervals whose lower endpoints are positive for its advantage over every competitor. This supports the principal STM lead across the tested strength grid, but it does not constitute a fully disjoint, task-wise optimized family ranking or a four-task Pareto analysis.

C Rate-profile optimization and comparison

The rate-profile analysis keeps the jump family fixed and changes only how local relaxation strength is distributed across sites. It therefore tests the second component of dissipator geometry independently of the jump-family comparison.

C.1 Within-family profile sweep

The profile sweep uses $N = 5$, mean local rate 1.5, $(h, \Delta t) = (0.5, 0.5)$, and a shorter protocol tailored to profile optimization. It uses 100 washout inputs, 100 optimization-training inputs, 150 validation inputs, and 200 test inputs. All static profiles have the same arithmetic mean. We compare a uniform reference with a random unequal profile and a static profile learned on validation data. A fourth model uses the input-adaptive profile $\gamma_i(s) = \mathrm{softplus}(a_i s + b_i)$. The adaptive profile is matched only in mean over the frozen input grid, so its instantaneous Frobenius strength may vary.

Static and adaptive profiles are optimized from seeded starts with fixed evaluation limits and stopping rules.

C.2 Does profile learning act the same way in every family?

A separate 2×2 check crosses local versus collective relaxation with uniform versus learned profiles. Figure 4(b) displays the 24 paired trajectories and their mean changes. Every candidate is normalized to the same $\mathcal{B}$, and exact continuous-input evolution replaces the dense approximation. For local relaxation, learning redistributes diagonal sitewise rates. For collective relaxation, it changes the coefficient vector $\boldsymbol{c}$ and thereby moves the supported one-dimensional Kossakowski direction while preserving rank one. The axes are therefore operational and interacting design coordinates rather than a globally orthogonal factorization. Inference uses simultaneous coverage over the eleven declared contrasts.

D Finite-sampling estimators and inference

The finite-sampling analysis asks whether the exact-expectation ordering remains visible when Pauli features must be estimated from a finite number of preparations. It compares two estimators at the same nominal preparation count, rather than treating shot noise as independent additive noise on every feature.

D.1 Two estimators at equal preparation count

At $N = 5$, write $N_{\text{prep}} = 45q$. The independent estimator spends q shots on each of the 45 observables. The grouped estimator instead spends $15q$ shots in each global X, Y, and Z product basis and reuses each measured bit string for all compatible one- and two-qubit features. Table 4 emphasizes the methodological difference without repeating the budget grid.

Table 4: **Finite-sampling estimators at equal nominal preparation count.** Grouping changes how measurement records are reused and therefore preserves correlations between compatible Pauli estimates.

	independent observables	grouped Pauli settings
recorded data	one binomial estimate per feature	full bit strings in X, Y, and Z bases
record reuse	none across features	one record informs all compatible features
covariance	omitted by construction	retained within each setting
modeled scope	projection noise	projection noise with grouping, without hardware noise

D.2 Training and simultaneous inference

Six non-dephasing designs are evaluated on 20 paired reservoir and measurement instances under both estimators. Validation selects the readout penalty from the predeclared grid before a train-plus-validation refit. We test $q = 64 \times 4^j$ for $j = 0, \ldots, 6$, so both estimators receive $45q$ preparations per step. Finite-budget intervals are simultaneous over every declared design-pair, estimator, and budget contrast. The exact-expectation endpoint is displayed separately.

The curves in Figure 5(a,b) should therefore be read as a resolution study. A design is distinguished from local relaxation only when the simultaneous interval excludes zero, not merely when its mean is larger. The model does not include acquisition latency, device drift, or gate and readout errors.

E Dynamical interpretation: support and limits

The following diagnostics test whether the observed response is consistent with the shared-channel interpretation without claiming a unique microscopic cause.

E.1 Slow dynamics that remain visible to the readout

For eight $N = 5$ local–collective lineages, the slowest right modes at five constant inputs are projected onto the 45-Pauli readout span. Collective relaxation retains slower modes with larger overlap. The paired differences are $0.338\,[0.239, 0.436]$ in decay rate and $0.104\,[0.028, 0.180]$ in readout-weighted retention. Because this omits excitation weights and switched propagator products, it supports but does not uniquely identify the coupling-geometry picture.

E.2 Concentrated response under switched input

Starting from synchronized trajectories, a complementary diagnostic applies small input perturbations and follows the 45-feature response over 20 lags. Collective relaxation concentrates this response in fewer observable patterns than the matched local process. Its effective rank is lower by $1.270\,[0.959, 1.580]$, while the leading-pattern energy fraction is higher by $0.285\,[0.243, 0.327]$. The switched-input result supports the same picture. It is an empirical response kernel rather than a complete modal decomposition.

E.3 A gradual path from local to collective coupling

To determine whether the endpoint contrast emerges along a continuous structural path, we interpolate the dissipators at fixed Frobenius weight as $\mathcal{D}_\alpha = (1-\alpha)\mathcal{D}_{\text{local}} + \alpha\mathcal{D}_{\text{collective}}$, with $0 \le \alpha \le 1$. Interior points are ranked on separate gap-only reservoirs before fresh task streams are opened. Increasing the collective component slows midpoint relaxation and raises STM at $N = 4$ and 5. The selected interior point beats local relaxation in every fresh pair. This links the endpoint result to a continuous within-family deformation, not to a general selector for unrelated processes.

E.4 Why driven dephasing forgets uniformly

The near-zero STM of the dephasing control admits a direct contraction argument under the connectivity assumptions of the primary model.

Proposition 1 (Uniform contraction of the driven dephasing model)**.** *Let $L_i = \sqrt{\gamma_i} Z_i$ with every $\gamma_i > 0$, let $\Phi_s = \exp(\mathcal{L}_s \Delta t)$ with $\Delta t > 0$, and suppose the nonzero single-spin-flip matrix elements of $H(s)$ connect the computational-basis hypercube for every $s \in [0, 1]$. Then there is a $\kappa < 1$ such that*

$$\sup_{s\in[0,1]} \|\Phi_s|_{\mathrm{Tr}=0}\|_{2\to 2} \le \kappa.$$

Here $\|X\|_2 = (\mathrm{Tr}\, X^\dagger X)^{1/2}$. Hence $\|\rho_n - I/2^N\|_2 \le \kappa^n \|\rho_0 - I/2^N\|_2$ for any switched input sequence and initial state.

Proof. For $X_t = \exp(t\mathcal{L}_s)X$, the Hamiltonian commutator is skew-adjoint in the Hilbert–Schmidt inner product, while the Hermitian-unitary jumps give $\frac{\mathrm{d}}{\mathrm{d}t}\|X_t\|_2^2 = -\sum_i \gamma_i \|[Z_i, X_t]\|_2^2 \le 0$. Equality at Δt would make the norm constant, so every $[Z_i, X_t]$ vanishes and X_t remains diagonal. Its derivative can remain diagonal only if $[H(s), X_t] = 0$. Hypercube connectivity then makes X_t scalar. Every nonzero traceless operator therefore contracts strictly. Attainment in finite dimension, continuity in s, and compactness of $[0, 1]$ give one common $\kappa < 1$. □

State differences are traceless, so the bound applies to every switched sequence and all $s \in [0, 1]$. It explains the dephasing zeros in Table 2 for the connected primary model, not arbitrary unital dissipators.

F Orientation within matched rank-one channels

To vary the supported collective direction while holding coarse channel invariants fixed, we use

$$c_i(f) = \exp[2\pi \mathrm{i} f(i-1)/5], \qquad f \in \{0, 0.25, 0.5, 0.75, 1\}.$$

Here $f = 0$ is the principal equal-phase channel and $f = 1$ is orthogonal to the uniform direction. Every condition has Kossakowski rank one, spectrum $(5,0,0,0,0)$, unit diagonal, $|c_i| = 1$, magnitude IPR $1/5$, and $\mathcal{B} = 80$; four frozen phase-scrambled controls provide additional zero-overlap directions.

The confirmation uses 32 fresh paired reservoirs and excludes the pilot. All nine conditions share the Hamiltonian, input, target, split, and readout within each pair. After an 800-input washout, the unchanged protocol uses 600 training and 400 untouched test inputs, delays 1–20, 45 Pauli features, and fixed ridge 10^{-8}. Four-state audits pass for all conditions on the first eight lineages. Figure 9 summarizes the intervention.

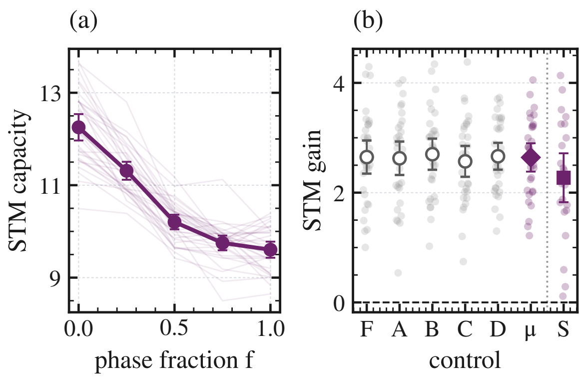

Figure 9: **Accessible memory depends on the matched rank-one direction.** *(a)* STM along the prespecified $N = 5$ phase path. *(b)* Equal-phase-minus-control effects for its five zero-overlap directions and their within-pair mean, followed by the separate $N = 6$ real sign-balanced replication (F, A–D, μ, and S on the axis). Faint marks show paired values; large marks and bars show means and 95% t intervals (32 and 24 pairs, respectively).

The equal-phase mean is 12.253, compared with 9.604 at the Fourier endpoint. Their paired difference is $2.648\,[2.347, 2.950]$, with 32/32 wins and prespecified two-sided sign-flip $p = 1.0 \times 10^{-5}$. Against the within-pair mean of all five zero-overlap directions, the gated effect is $2.642\,[2.383, 2.900]$, again with 32/32 wins and $p = 1.0 \times 10^{-5}$. The ordered path and validation-selected-ridge sensitivity give the same conclusion.

Real sign-balanced replication at a second finite size. We independently repeat the endpoint intervention in 24 separately generated $N = 6$ pairs using $c_+ = (1,1,1,1,1,1)$ and $c_\perp = (1,1,1,-1,-1,-1)$, with $c_+^\dagger c_\perp = 0$. The channels match in jump count, rank-one Kossakowski spectrum $\{6,0,0,0,0,0\}$, trace 6, diagonal weights, coefficient magnitudes, and assigned weight $\mathcal{B} = 192$. Within each pair, the processor, input, readout, data, and ridge rule are identical. All 24 ground/mixed-state audits pass at washout 800; the first six also pass the full four-state audit.

Equal-phase STM is 13.626, versus 11.354 for the orthogonal direction: the paired gain is $2.271\,[1.826, 2.717]$, with 24/24 wins and exact two-sided sign-test $p = 1.19 \times 10^{-7}$. Every delay-wise mean difference is positive. The response is more concentrated under equal phase, but those diagnostic changes are not detectably associated with the STM gain and are not identified mediators. Thus matched rank, spectrum, trace, diagonal weights, coefficient magnitudes, and assigned weight do not determine accessible memory: orientation relative to the fixed processor is itself a computational coordinate. This is a finite-size result, not a scaling law, universal optimum, or unique microscopic mechanism.